

# Project Design Phase

## Solution Architecture

|               |                                            |
|---------------|--------------------------------------------|
| Date          | 16 September 2026                          |
| Team ID       | PNT2022TMIDxxxxxx                          |
| Project Name  | Car Rental Request in ServiceNow           |
| Technology    | ServiceNow – Service Catalog and Workflows |
| Maximum Marks | 4 Marks                                    |

### Solution Architecture:

The solution architecture for the Car Rental Request system bridges the rental-request business need with ServiceNow capabilities. It describes how the Service Catalog, request records, workflow automation, availability checking, manager approval, fulfillment and exception handling work together.

### Architecture Goals

- Provide a standardized Service Catalog entry for submitting car rental requests.
- Automate the request lifecycle using ServiceNow Workflow / Flow Designer.
- Check configured car availability before proceeding with fulfillment.
- Route the request to the configured manager approval step.
- Create a fulfillment Catalog Task for the Car Rental / Fleet team where configured.
- Handle an unavailable vehicle through the configured exception / incident process.
- Maintain centralized ServiceNow records for request, approval, task and incident tracking.

### Solution Architecture Diagram

|                                                                         |                                                                               |                                                                                   |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>REQUESTER</b><br>Employee / User<br>Submits rental details           | <b>SERVICE CATALOG</b><br>Car Rental Request<br>Catalog variables             | <b>WORKFLOW / FLOW DESIGNER</b><br>Automated processing<br>Conditions and actions |
| <b>REQUEST RECORD</b><br>Request / Requested Item<br>Status tracking    | <b>AVAILABILITY CHECK</b><br>Car Inventory<br>Car Type + Location + Available | <b>MANAGER APPROVAL</b><br>Approve / Reject<br>Approval status                    |
| <b>FULFILLMENT</b><br>Catalog Task [sc_task]<br>Car Rental / Fleet team | <b>EXCEPTION HANDLING</b><br>Unavailable car<br>Incident when configured      | <b>FINAL STATUS</b><br>Request outcome<br>Notifications where configured          |

**Primary flow:** Requester → Service Catalog → Workflow / Flow Designer → Availability Check → Manager Approval → Fulfillment or Rejection → Final Status. If the car is unavailable, the configured exception path can create an Incident.

### Architecture Components

| Component                | Purpose in the Project                                                                  |
|--------------------------|-----------------------------------------------------------------------------------------|
| Service Catalog          | Provides the Car Rental Request form and captures required rental information.          |
| Catalog Variables        | Car Type, Pickup Location, Drop Location, Requested Date and Justification.             |
| Workflow / Flow Designer | Controls request processing, conditions, approvals, tasks and configured notifications. |
| Car Inventory            | Stores/configures vehicle information used for the availability check.                  |

| Component                 | Purpose in the Project                                                                         |
|---------------------------|------------------------------------------------------------------------------------------------|
| Manager Approval          | Provides the approval or rejection decision before fulfillment.                                |
| Catalog Task [sc_task]    | Supports fulfillment by assigning the request to the Car Rental / Fleet team where configured. |
| Business Rule             | Can prevent car assignment when Availability is not Available.                                 |
| Incident                  | Used by the configured unavailable-car exception path for follow-up.                           |
| Request / Related Records | Centralized tracking of request status, approval, task and incident records.                   |

# Solution Architecture – Process and Data Flow

## Car Rental Request in ServiceNow

### Process Flow

| Step | Stage                     | ServiceNow Processing                                                                                                                 |
|------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Request Submission</b> | User opens the Car Rental Request catalog item and enters car type, pickup location, drop location, requested date and justification. |
| 2    | <b>Request Creation</b>   | ServiceNow creates the request / requested item record and starts the configured workflow.                                            |
| 3    | <b>Availability Check</b> | The workflow checks configured Car Inventory records for a matching car type, pickup location and Available status.                   |
| 4    | <b>Approval</b>           | The request is routed to the configured manager approval step. The manager approves or rejects the request.                           |
| 5    | <b>Fulfillment</b>        | For an approved and available request, the configured fulfillment task can be created and assigned to the Car Rental / Fleet team.    |
| 6    | <b>Exception</b>          | If the requested car is unavailable, the configured exception path can create an Incident for follow-up.                              |
| 7    | <b>Completion</b>         | The request status is updated through the configured process and related records provide the final outcome.                           |

### Key Requirements Addressed

- **Standardized intake:** one Service Catalog item for the Car Rental Request.
- **Automation:** workflow / Flow Designer controls the request lifecycle.
- **Availability:** configured Car Inventory records are evaluated before fulfillment.
- **Approval:** manager approval determines whether the request can proceed.
- **Fulfillment:** a Catalog Task can be assigned to the Car Rental / Fleet team.
- **Exception handling:** unavailable-car cases can follow the configured Incident path.
- **Control:** a Business Rule can prevent assignment when a vehicle is not available.
- **Tracking:** ServiceNow records provide a centralized view of request progress and outcome.

### Architecture Benefits

The architecture provides a single ServiceNow-based process for car rental requests. It separates the user request interface from automated processing, approval and fulfillment, while keeping related records within the ServiceNow platform.

### Template Alignment

The uploaded template is a Project Design Phase document titled "Solution Architecture". It describes architecture as the bridge between business problems and technology solutions and emphasizes technology selection, software structure and behavior, features, requirements and delivery specifications.

■filecite■turn18file0■L2-L17■ The sample also contains an architecture/data-flow diagram; this document replaces that example with a project-specific ServiceNow architecture.